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Unique 3D metamaterial split ring resonator for wireless communication with high negative refractive index and for medium ratio

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ABSTRACT

This article presents the unique 3D metamaterial split ring resonator (3D-MTM SRR) model with high negative refractive index and effective medium ratio at 2.45 GHz frequency. It is integrated with two layers: one square bracket structure printed on the back side of the substrate and another dual split ring with an air gap placed above the substrate. Both structures are coupled by metallic vias, which serve as connecting vias to create the 3D-MTM SRR model. Simulation and experiment results shows good agreement for high negative refractive index and effective medium ratio of −17 and 6.16, respectively. To fully comprehend the resonance phenomena of the proposed 3D-MTM model, an analogous circuit analysis is carried out. In addition, to validate the MTM activities, array topologies based on 3D-MTM cells are analyzed. Thus the unique properties of 3D-MTM SRR allow to develop the high-performance RF devices for wireless applications.

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KEYWORDS
3D-Metamaterial; effective permittivity; effective permeability; negative refractive index; split ring resonator.

1. Introduction

Wireless communication has been more popular among researchers over the last few decades. To develop modern wireless technologies, several scientific researchers frequently use a metamaterial (MTM), which has unique electromagnetic characteristics. In general, MTM is an artificial material that has unique effective medium characteristics [1–5]. The MTM properties such as effective parameters (permittivity and permeability) and refractive index are achieved by designed structures. These characteristics/designs are suitable for a wide range of application [6–10].

Karim et al. [11] proposed a sandwiched metamaterial with several square split ring resonators and a set of wires that exhibits left-hand properties in the microwave frequency band. Ramachandran et al. in [12] proposed a multiresonance square shaped MTM structure with an effective medium ratio of 15.873 for L- and C-band wireless applications. Although the proposed unconventional MTM designs in this research offer great features, they are not widely used. Afsar et al. proposed a triple-band epsilon-negative (ENG) metamaterial properties based on a split ring resonator [13]. This structure exhibits a negative

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permittivity MTM unit cell may be used for various wireless applications. On the other hand, a greek-key-shaped MTM structure was demonstrated by Zarghooni et al. [14]. Here, the MTM structure shows the multiple bands behavior in 1–5 GHz. The physical dimensions is quite small and have a near-zero refractive index. In another research work, a new substrate material integrated with MTM design was proposed by Elwi et al. [15]. Here, a very thin antenna structure based on an MTM dipole was printed on substrate material. A previous study has revealed that authors concentrated their attention on frequency bands. The MTM absorber has become popular in the scientific world, in addition to its antenna use.

The non-symmetric artificial MTM design with negative refractive index characteristics was proposed by Harbinder et al. [16]. This dual layer MTM shows the negative permittivity and permeability up to 10 GHz and a positive permittivity and permeability above 10 GHz. Borazjani et al. [17] recently proposed a simple MTM unit cell with negative permittivity and array structure with reduced mutual coupling in the frequency range 9.2–11.7 GHz. Meanwhile, Cheng et al. proposed an anisotropic MTM composed of a bi-layered disk-SRR array and sandwiched sub-wavelength metal grating. This MTM array was used for polarization converters with more than 99% efficiency at 4.5 GHz and 7.9 GHz [18]. Besides MTM absorbers, an antenna application is one of the emerging topics in the research community. A patch antenna is presented by Zhou et.al [19], where a metasurface was used for wideband high efficiency cross-polarization reflection. An array of MTM unit cells was used to increase the bandwidth and gain [20]. The mentioned article shows the limitation of negative refractive index and effective medium ratio (EMR) for wireless communication systems.

Recently, researchers show interest in 3D metamaterial (3D-MTM) structures because of their ability to control wave propagation in the medium, carry power in a controlled manner, small size, and broad-bandwidth for the modern communication system. The 3D-MTM structure was designed using a variety of fabrication techniques, including copper conductive filament was proposed in [21] and LTCC technology for microwave applications was developed by Vardan et al. in [22]. In [23], Viskadourakis et al. proposed the fused filament fabrication technology to create the 3D printed cut-wire MTM structures, which harvests energy at frequencies of about 2.4 GHz. These procedures are expensive and complex, but they can be solved by transforming the 2D structure into a 3D shape. In [24] proposed SRR-based variant 3D geometry of unit cells to exhibit DNG MTM property for tuning frequency band thereby increasing bandwidth. Meanwhile, Choi et al. [25] proposed the miniaturized 3D-MTM structure. Here, the metallic via is connected to the SRR to get the 3D-MTM resonance structure. Hence, the electrical length is increased and the negative permeability and negative refractive index were achieved. The detailed parametric analysis and advantage of SRR-based 3D-MTM were discussed in [26] by Husna et al. The above-mentioned articles have limitations in negative refractive index, complex MTM structures, and more surface wave loss.

To overcome such challenges in MTM structure, a unique 3D-MTM SRR unit cell and array with a high negative refractive index and an effective medium ratio have been proposed in this article at 2.45 GHz frequency for modern wireless communication systems. The proposed design is based on two layers: a square bracket structure and a dual split ring, which is connected with metallic vias to form the 3D-MTM SRR model. The metallic via interacts with the square bracket structure and the dual split ring reduces the surface wave and refracted

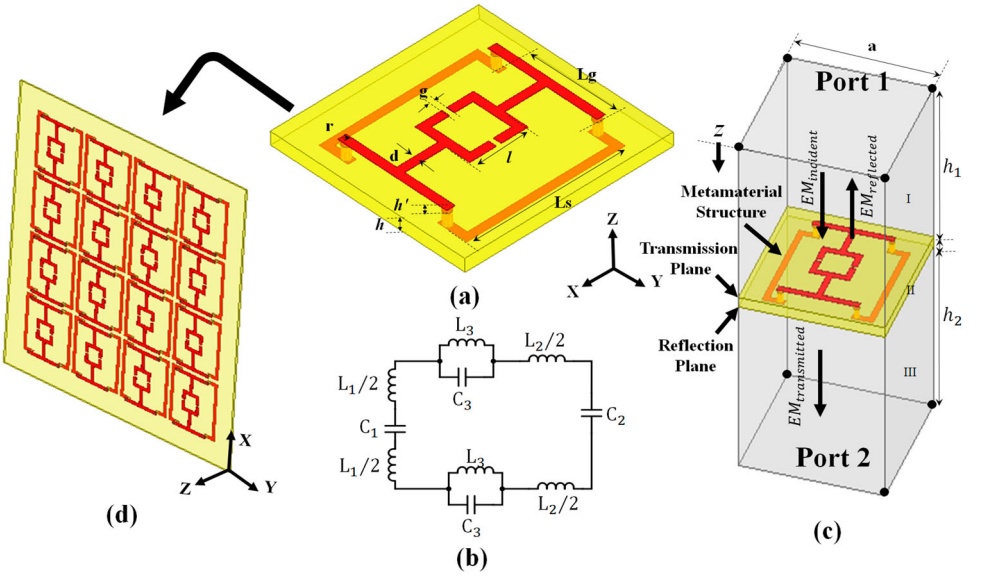


Figure 1. Geometry of proposed 3D-MTM SRR model (a) 3D-MTM SRR unit cell, (b) equivalent circuit model, (c) 3D-MTM SRR model simulation setup, (d) 3D MTM SRR unit cell array structure.

wave in the 3D-MTM SRR structure. As a result, negative permeability and the negative refractive index have been achieved at 2.45 GHz. Besides, the effective parameters for the proposed 3D-MTM SRR and array structure show a high negative refractive index and effective medium ratio. An analogous circuit model has been proposed to anticipate the effects of structure parameters, which is then tested through simulation and measurement with the satisfactory agreement.

2. Proposed 3D MTM SRR model

2.1. Geometry and its equivalent circuit model

Figure 1 shows the proposed 3D MTM SRR model associated with the equivalent circuit model and dimensions are listed in Table 1. The proposed SRR is combined with two layers: one square bracket structure printed on the bottom of the substrate and another dual split ring is placed above the substrate with an air gap (h') as shown in Figure 1(a). The air gap acts as a capacitive coupling between the dual split ring and substrate. The square bracket structure is printed on substrate Rogers 5880 ($\varepsilon_r = 2.2$, $\tan\delta = 0.0009$) with height (h). The metallic vias of radius (r) are connected with square bracket structure and dual split ring. The initial dimensions were calculated using general subwavelength principle as $\lambda/10$ at 2.45 GHz frequency. Hence, the resonant frequency can be calculated from the equivalent circuit model shown in Figure 1(b) is written as

$$f = \frac{1}{2\pi\sqrt{L_1C_1L_2C_2}} \tag{1}$$

The effective inductance and capacitance of the dual split ring and square bracket structure are represented by L_1, L_2 and C_1, C_2, respectively. As illustrated in Figure 1(c), the

Table 1. Dimensions (in mm) of proposed 3D-MTM SRR model.

Parameters	L_s	L_g	l	d	g	r	h	h'
Values	20	14	7	1	0.65	0.5	0.51	0.5

proposed 3D-MTM SRR unit cell model activates by a perpendicularly incident polarized wave causing voltage to be introduced in the copper metal gaps or parallel metal strip. It can be approximated with the development of parallel plate capacitance principle. As a result, the capacitor's discharging path can be used to represent inductance. Therefore, the electric resonance has been generated by combining split and electric fields. Similarly, the magnetic resonance is generated by combining metal loop and magnetic field. The below equations represent the formation of capacitance and inductance for the proposed 3D-MTM SRR model [1].

For capacitance,

$$C_1(F) = \varepsilon_0 \varepsilon_r \frac{A'}{l} \tag{2}$$

and

$$C_2(F) = \varepsilon_0 \varepsilon_r \frac{A''}{L_s} \tag{3}$$

where free space permittivity is ε_0, relative permittivity is ε_r. The l and L_s are the length of square bracket and dual ring, respectively. Similarly, the cross section area and length of a dual split ring and square bracket structure are A', A'' and l, Ls. And for inductance,

$$L_1(nH) = 2 \times 10^{-4} l \left[\ln \frac{l}{d + t'} + 1.193 + 0.02235 \frac{d + t}{l} \right] k_g \tag{4}$$

and

$$L_2(nH) = 2 \times 10^{-4} L_s \left[\ln \frac{L_s}{d + t} + 1.193 + 0.02235 \frac{d + t'}{L_s} \right] k_g \tag{5}$$

where d is the width of the strip for the 3D-MTM SRR model. k_g is the correction factor, which is related to the length (l and L_s) and height (h) of substrate. The $t = 0.508$ mm and $t' = 0.036$ mm are the copper thickness of dual split ring and square bracket structure, respectively. As seen from the equivalent circuit model, the metallic via has self-inductance (L_3) and capacitance (C_3), which is serially connected to the inductance of dual split ring and square bracket structure. When physically evaluated, it is expressed as [1]

$$L_3 = \frac{\mu_0 \mu_r}{\pi} \cosh \left(\frac{h + h'}{r} \right) t' \tag{6}$$

and

$$C_3 = \frac{\varepsilon_0 \varepsilon_r A}{h + h'} \tag{7}$$

where μ_0 is the free space permeability, μ_r is the substrate permeability. $h + h'$ is the height of the metallic via, r is the radius of metallic via and $A = 2rh$ is the cross section area of the metallic via. Table 2 shows optimized values for the equivalent circuit model and these values are used to plot the S-parameter in HFSS circuit simulation tool.

Table 2. Extracted lumped values of proposed 3D-MTM SRR unit cell model.

Parameters	L_1 (nH)	L_2 (nH)	L_3 (pF)	C_1 (pF)	C_2 (pF)	C_3 (pF)
Values	0.1776	6.122	3.516	5.049	0.8146	0.2146

2.2. 3D MTM performance setup

The performance of the proposed model is determined using the NRW method [11]. The unit cell is placed inside the waveguide in simulation as shown in Figure 1(c). Two waveguide ports are used for EM wave propagation along the *z*-axis. The perfect electric and magnetic field boundary is applied along the *x*-axis and *y*-axis. Hence, the electric fields of the reflected and transmitted wave are integrated in reflection plane and transmission plane, respectively in three different sections E_I, E_{II} and E_{III}. It is written as

$$E_I = e^{(-\gamma_0 h_1)} + B_1 e^{\gamma_0 h_1} \tag{8}$$

$$E_{II} = B_2 e^{(-\gamma(h_1+h+h'))} + B_3 e^{\gamma(h_1+h+h')} \tag{9}$$

$$E_{III} = B_4 e^{(-\gamma_0(h+h'+h_2))} \tag{10}$$

where $\gamma = j[(\omega/c)^2 - (2\pi/\lambda_c)^2]^{1/2}$ and $\gamma_0 = j[(\omega\mu_r\varepsilon_r/c)^2 - (2\pi/\lambda_c)^2]^{1/2}$ are the propagation constant for air and unit cell respectively. Here, c is the speed of light, ω is the angular frequency and $\lambda_c = 2a$ is the cut-off wavelength and a is the width of wave guide. The constant $B_i (i = 1,2,3,4)$ can be determine from the boundary condition of electric and magnetic field. The total height of the wave guide is $h_{total} = h_1 + h + h' + h_2$. Here, h_1 and h_2 are the height of the receptive port and h, h' are the height of the substrate and air gap of 3D-MTM unit cell model. Therefore, the scattering parameters of these two ports are calculated as

$$S_{11} = P_1^2 \left[\frac{\Gamma(1-T^2)}{1-\Gamma^2 T^2}\right] \tag{11}$$

$$S_{22} = P_2^2 \left[\frac{\Gamma(1-T^2)}{1-\Gamma^2 T^2}\right] \tag{12}$$

$$S_{12} = S_{21} = P_1 P_2 \left[\frac{\Gamma(1-T^2)}{1-\Gamma^2 T^2}\right] \tag{13}$$

where P_1 and P_2 are the two-port reference plane transformations and written as [$P_i = e^{-\gamma_0 h_i}$ (i = 1, 2)]. The Γ and T are the reflection coefficient and transmission coefficient for the proposed 3D-MTM SRR model are calculated as $\Gamma = [(\gamma_0/\mu_0) - (\gamma - \mu)][(\gamma_0/\mu_0) + (\gamma - \mu)]^{-1/2}$ and $T = e^{-\gamma L}$. Both are used for scattering parameters calculation equations (11)–(13). The S-parameters are simulated and measured for frequencies ranging from 1 to 4 GHz. Figure 2(a) shows the fabricated prototype of the 3D-MTM SRR unit cell positioned between two waveguide ports for measurement and its experimental setup validates the proposed design. Figure 2(b) shows S-parameter values of the equivalent circuit, simulated and measured for the proposed model. The magnitude of the calculated s-parameters matches the measurement quite well. The tiny difference between the simulated and measured values was observed due to the measurement setup's alignment. The scattering parameters have a bandgap between 2.36 and 2.53 GHz (resonate at 2.45 GHz)

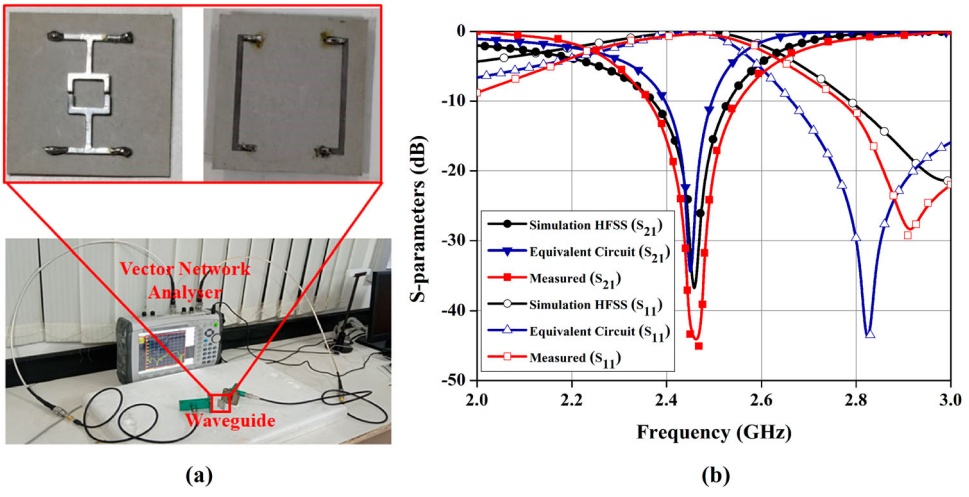


Figure 2. Proposed 3D MTM SRR unit cell (a) fabricated prototype and measurement setup and (b) S-parameters.

and shows negative material properties for the proposed 3D-MTM SRR model. Figure 3 shows the excited surface current distribution at 2.45 GHz (*xy*-plane). The color represents the current intensity, while the arrows indicate the current direction. Current flows in the opposite direction through metallic vias from a dual split ring to a square bracket structure. Because the opposing currents are equal, the fields are internally cancelled, resulting in a stop band. This instance shows strong interaction of fields between the square bracket structure and the dual split ring, resulting in a negative permeability and negative refractive index.

Figures 4(a)–(d) show the simulated and measured scattering parameters phase, permittivity, permeability, and refractive index for the proposed model. Figure 4(a) shows the simulated and measured S-parameter phase magnitude. These parameters are well matched with S-parameters magnitude (see Figure 2b) and show the 3D-MTM SRR properties at 2.45 GHz frequency. The effective medium parameters are calculated for 3D-MTM SRR material properties as follows:

The complex permittivity

$$\mu_r = \frac{2}{jk_0h}\left[\frac{1-H_2}{1+H_2}\right] \tag{14}$$

the complex permeability

$$\varepsilon_r = \frac{2}{jk_0h}\left[\frac{1-H_1}{1+H_1}\right] \tag{15}$$

and refractive index

$$\eta = \sqrt{\varepsilon_r \mu_r} \tag{16}$$

where $H_1 = (S_{21} + S_{11})$ and $H_2 = (S_{21} - S_{11})$, k_0 the wave number and η is the refractive index. Figures 4(b) and (c) present the simulated and measured effective relative permittivity and permeability (real and imaginary) over the frequency range of 2.36–2.53 GHz for

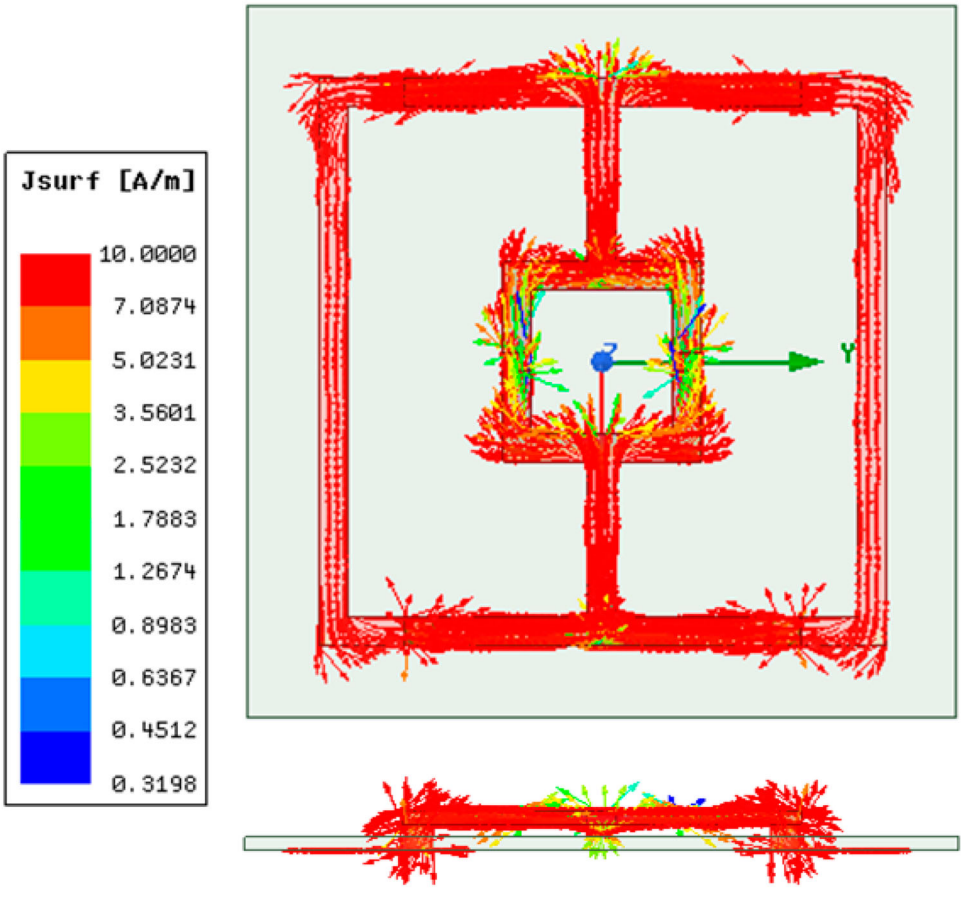


Figure 3. Surface current distribution of the proposed 3D-MTM SRR model at 2.45 GHz.

the proposed model, extracted from the S-parameters. The effective permittivity and effective permeability are found to be negative and positive, respectively in the 2.36–2.53 GHz band. It can be seen from the preceding statement that the effective permittivity is the positive value of 40 and the effective permeability is the negative value of −35. Figure 4(d) shows the effective refractive index for the proposed 3D-MTM SRR model. At the frequency of 2.45 GHz, the effective refractive index is found to be high negative value of −17. It is observed that these values clearly indicate the proposed 3D MTM behaves as a μ-negative MTM. To enhance the performance of microwave devices in wireless communication system, it is necessary to design the array of proposed 3D-MTM SRR model. The next section describes the detailed investigation on array model for the proposed 3D-MTM SRR.

3. 3D MTM array performance results

3.1. Performance of different array structure

To check the performance in the array configuration for the proposed 3D-MTM SRR unit cell model, three different types of configuration (1 × 2, 2 × 1 and 2 × 2) have been taken

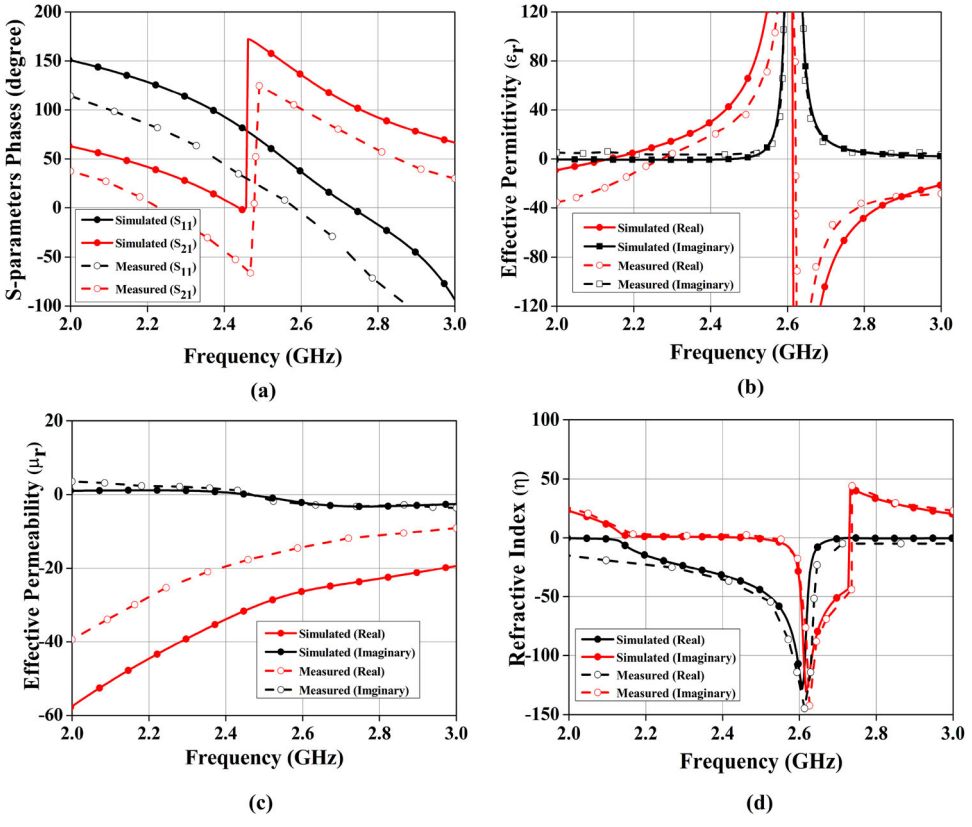


Figure 4. Simulated and measured effective parameters of proposed 3D-MTM SRR unit cell model: (a) S-parameter phase magnitude, (b) effective permittivity, (c) effective permeability, and (d) refractive index.

into consideration. The geometrical configurations of 1×2, 2×1 and 2×2 are shown in Figure 5 and the dimensions are $25 \times 50\,\text{mm}^2$, $50 \times 25\ \text{mm}^2$, and $50 \times 50\,\text{mm}^2$, respectively. Figure 5(a) shows the 1×2 array of the 3D-MTM SRR unit cell, which is created by connecting two unit cells horizontally. Figure 5(b) shows the 2×1 array configuration with two unit cells joint horizontally. Similarly, Figure 5(c) depicts the 2×2 array of the 3D-MTM SRR unit cell, which was created by joining two unit cells horizontally and vertically. Figure 6(a) presents S_{11} and S_{21} parameters of the different array configuration. In all configurations, 2.45 GHz resonant frequency is obtained. Effective medium parameters of the array were calculated using the same 3D-MTM SRR unit cell methodology. Figures 6(b)–(d) show the effective permittivity, effective permeability, and refractive index for the $1 \times 2, 2 \times 1$, and 2×2. Figure 6(b) shows the effective permittivity for the different array configuration. It is observed that the real value in all configurations is positive. Figure 6(c) shows the effective permeability for the different array configuration. It is observed that the real value in all the configuration is negative. The above observation indicates the MTM properties for the array configuration at 2.45 GHz frequency. Figure 6(d) shows the refractive index for the different array configuration. This is extracted from the effective permittivity and permeability. The refractive index at the resonant frequency of 2.45 GHz has a high negative value. It concludes that these values are showing the MTM behavior

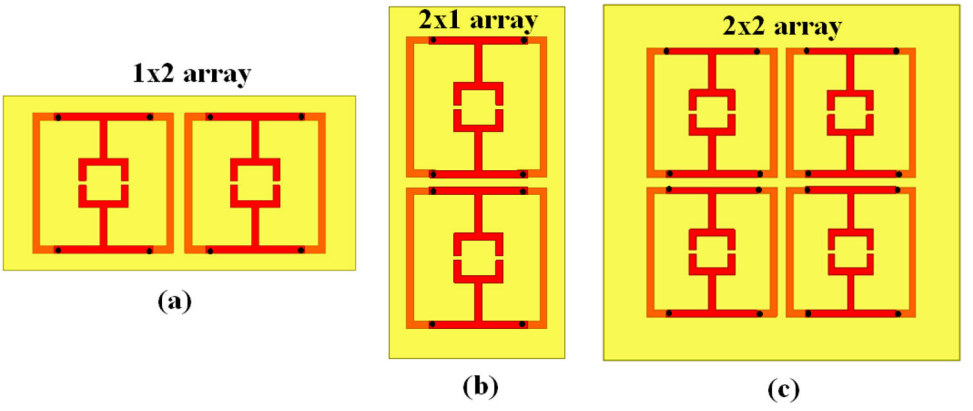


Figure 5. Different array configuration for the proposed 3D-MTM SRR unit cell model: (a) 1 × 2 array, (b) 2 × 1 array, and (c) 2 × 2 array.

for the proposed geometry. Hence, the structure can be used to design the large array configuration. The next section describes the large 4 × 4 array configuration for the proposed 3D-MTM SRR model.

3.2. Experimental verification of 3D MTM array:

This section explores the behavior of the proposed 3D-MTM SRR unit cell model when used in the large array configuration. Hence, the single unit cell was then extended to 4 × 4 array structure, as shown in Figure 1(d). The distance between the two cells is 1.5 mm. The overall dimensions of the 4 × 4 array structures are 100 × 100 mm^2. The array performance is investigated using a simulation setup that is set up in the same way as a unit cell, and surface current distribution and simulated S-parameters are obtained. Figure 7(a) shows the surface current distribution of the proposed 4 × 4 array 3D MTM structure. It is found that the observation is the same as a unit cell and the surface current is equally distributed in the entire 3D-MTM SRR model. The proposed array structure was measured using a free space measurement system [21]. A fabricated photograph (inset) of array and measurement setup with the 3D-MTM SRR sample is presented in Figure 7(b). The results s-parameters for the proposed array antenna were validated using a measurement setup. A vector network analyzer (VNA), two standard horn antennas for transmitter and reception, a sample holder, and two coaxial cables to waveguide adapters comprise the measuring setup. The dimension of the horn antenna will be determined by the testing frequency range. The experiment is demonstrated in the S-band (1.72–2.6 GHz). The waveguide standard for S-band is WR 8, which has a cut-off frequency (f_c) of 1.372 GHz and a dimension of 109 × 54 mm^2. The gap between the horn antenna and the fabricated prototype needs to be estimated using the simulation model assumption. In a simulation, the distance between an array of unit cells and ports is about λ/4. Whereas in measurement the separation distance is approximated to 35 cm, which is greater than the far-field approximation of $2D^2/\lambda$. To determine the appropriate reflector in the incident wave, a copper plate is first placed between two horn antennas. Then for measurement, the fabricated prototype is mounted between the

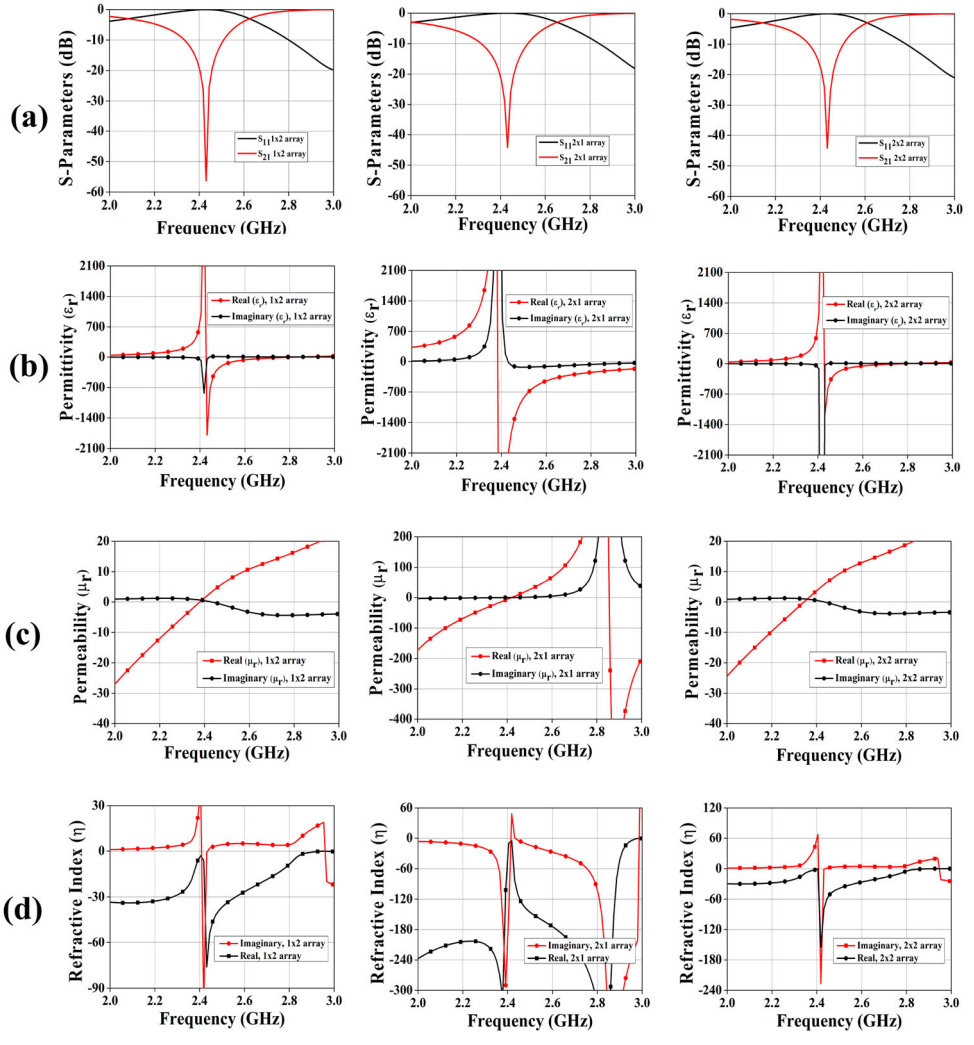


Figure 6. Simulated effective medium parameters for the proposed different 3D MTM SRR array configurations: (a) S-parameters, (b) effective permittivity, (c) effective permeability, and (d) refractive index.

horn antennas. In terms of polarization angle variation (ϕ), the wave propagates towards the *z*-directions, that's why normal occurrences occur.

The S-parameters, effective permittivity, effective permeability, and refractive index of the proposed 3D MTM array structure are shown in Figure 8(a–d). Figure 8(a) shows the simulated and measured transmission coefficient ($S_{21} = S_{12}$) for the proposed 4 × 4 3D-MTM array model, which is exhibited to resonate at 2.45 GHz frequency. The simulation frequency range was found to be 2.36 –2.53 GHz, whereas the measurement frequency range was found to be 2.32 –2.56 GHz. Even though the resonance frequency has shifted somewhat, there is good agreement between the simulated and measured results. Although the resonance frequency shifts due to fabrication mistake, measurement methodologies, or mutual interaction amongst array prototypes, the prototype still reaches the optimized

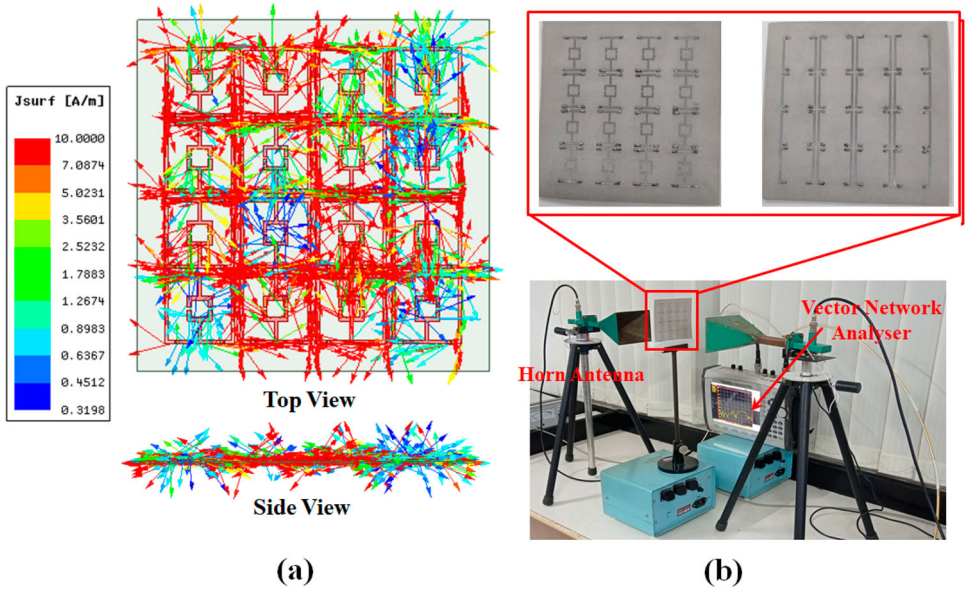


Figure 7. 4 × 4 3D MTM SRR array model: (a) surface current distribution and (b) fabricated prototype and measurement setup.

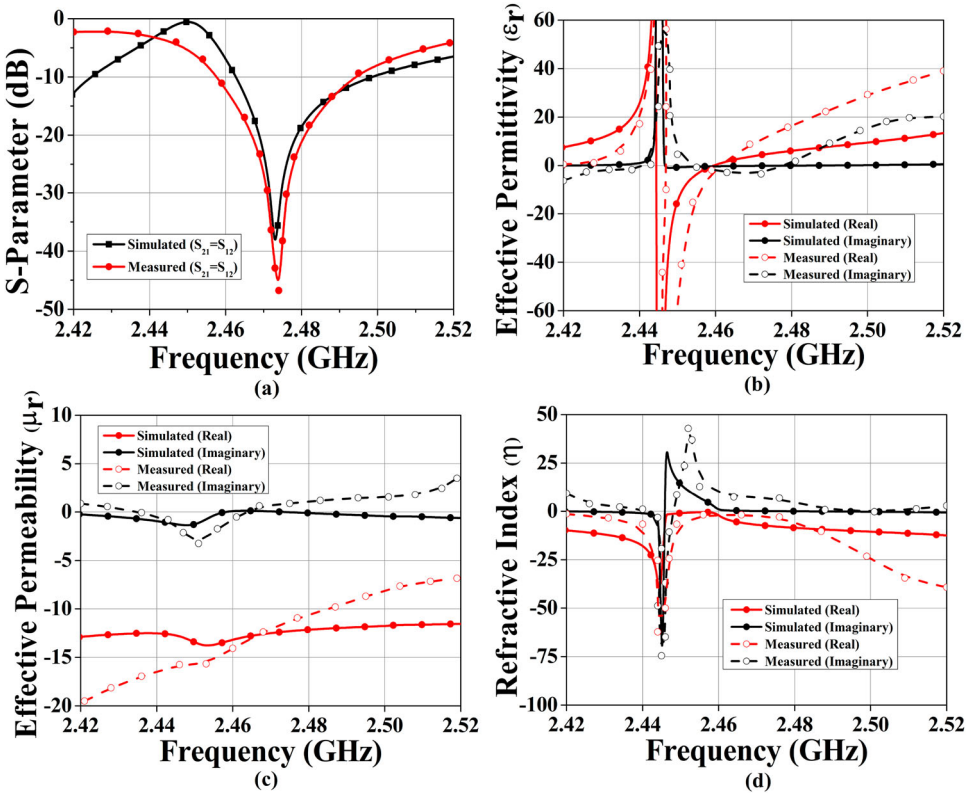


Figure 8. Simulated and measured effective parameters of proposed 3D-MTM SRR 4 × 4 array model: (a) S-parameters, (b) effective permittivity, (c) effective permeability, and (d) refractive index.

Table 3. Comparison between the proposed work and recent literature. NRI = Negative Refractive Index, EMR = Effective Medium Ratio.

Ref.	Unit cell shape	Size (mm^2)	Frequency (GHz)	MTM Type	NRI Index	EMR
[10] (2020)	2D SRR	20 × 20	8–18 GHz	Double -ve medium	–	–
[11] (2020)	2D SRR	15 × 15	1–6 GHz	MNG -ve medium	−13	15.973
[12] (2021)	2D SRR	9 × 9	2.9–3.7 GHz 9.5–10.1 GHz	ENG -ve medium	–	11.53
[15] (2019)	2D SRR	10 × 10	2.45 GHz 5.8 GHz	MNG -ve medium	3.75	–
[16] (2020)	2D SRR	9 × 7	9.2–11 GHz	MNG -ve medium	−1.4	0.2
[17] (2021)	2D CSRR	5 × 5	10 GHz 7.9 GHz	Double -ve medium	–	–
[23] (2021)	3D MTM	23 × 22	2.4 GHz	Eplison -ve medium	–	–
[25] (2017)	3D SRR	20 × 1	3.03 GHz	–	–	–
This work	3D MTM SRR	20 × 20	2.36–2.53 GHz	MNG -ve medium	−17	6.16

resonance frequency of 2.45 GHz. The permittivity (ε) and permeability (μ) have also been calculated for the proposed array structure from the simulated and measured S-parameters at 2.45 GHz frequency as shown in Figure 8(b) and (c). The simulated and measured values are well matched. It is observed that the measured real value of the ε is positive around 0.97 and the measured real value of the μ is negative around −3.18. Both the values prove the 3D-MTM SRR characteristic for the 4 × 4 3D-MTM array configuration. Besides Figure 8(d) presents the simulated and measured values of refractive index (η) at 2.45 GHz frequency, it is extracted from permittivity (ε) and permeability (μ). The refractive index for the proposed 4 × 4 3D-MTM array configuration is highly negative at a frequency 2.4697 GHz. Because of this property, the suggested 3D-MTM SRR array design can be used for RF microwave device in wireless communication applications. In addition, the effective medium ratio (EMR) is a crucial element in defining MTM characteristics. It shows the compactness of the MTM and calculated from the Equation (17) given below:

$$EMR = \frac{\lambda}{L} \tag{17}$$

where λ represents the wavelength and L represents the overall length of the proposed 3D-MTM SRR unit cell. The EMR of the proposed 3D-MTM SRR unit cell is 6.16 for 20 × 20 × 0.51 mm^3 dimension in the frequency of 2.45 GHz. Negative permeability and/or permittivity in the effective medium region may be shown by the unit cell if the EMR is greater than 4, which is the ideal value. Table 3 represents the comparison for proposed 3D-MTM SRR model and existing MTM. Finally, it is stated that the unique 3D MTM SRR unit cell and array design may be used to a single conventional or array microstrip antenna to cluster the incident radiation field and boost directivity. As a result, the unit number of antenna arrays might be lowered while maintaining the same gain, which is essential for decreasing antenna array design for future wireless communication applications.

4. Conclusion

A unique 3D-metamaterial split ring resonator (3D-MTM SRR) unit cell and its associated array layout with high negative permeability and negative refractive index are presented in this work. The proposed 3D-MTM SRR demonstrates the resonance of transmission coefficients covering 2.36–2.53 GHz and resonant at 2.45GHz, as performed for simulation and measurement. At resonance frequency, the high refractive index and EMR of −17 and 6.12 are attained for unit cell, respectively. A 4×4 3D MTM array was designed and tested. The unique properties of the proposed 3D-MTM SRR allow to develop the high-performance RF devices for future wireless applications.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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